

Assignment 2

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1 Introduction

This assignment investigates the robustness of different networks under progressive node removal attacks. Robustness is evaluated by monitoring the size of the giant component (GC) during node removal. A network is considered robust if the GC remains large even after many nodes are removed.

Three networks are analyzed:

- an Erdős–Rényi random graph (ER);
- a Barabási–Albert scale-free graph (BA);
- a network built from the MyAnimeList dataset.

Four attack strategies are applied:

- random node removal;
- highest degree node removal;
- highest PageRank node removal;
- highest betweenness node removal (approximate).

Finally, different edge-addition strategies are applied to the anime network in order to evaluate possible improvements in robustness.

2 Attack Methodology

The attack procedure is iterative:

1. compute the largest connected component;
2. select nodes according to the attack strategy;
3. remove the selected nodes;
4. recompute the largest component.

The robustness measure is:

$$\frac{G}{N}$$

where G is the size of the giant component after removal and N is the initial number of nodes.

The collapse point is defined as the moment when the giant component falls below 50% of its original size.

Technical Decisions

Several technical decisions were required to ensure correctness and computational feasibility:

- **Static attacks:** degree, PageRank and betweenness rankings are computed once on the initial graph and reused during the attack.
- **Approximate betweenness:** betweenness centrality is computed using sampling.
- **Batch removal:** for large networks, attacks remove 0.5% of the GC at each iteration.
- **Giant component tracking:** after each removal, only the largest connected component is considered.
- **Optimized graph construction:** the anime network is built using an inverted index to avoid $O(N^2)$ pairwise comparisons.
- **Similarity rule:** two anime are connected if they share at least two genres.

These decisions reduce computation time while preserving correctness and ensuring that attack curves are not distorted.

3 Experiments on Synthetic Networks

3.1 Erdős–Rényi Network

The ER graph ($N = 1000$, $p = 0.01$) shows homogeneous robustness. Random failures and targeted attacks produce identical behavior because the degree distribution is relatively uniform.

Collapse points:

- Random: 500 nodes removed (50.0%),
- Degree: 500 nodes removed (50.0%),
- PageRank: 500 nodes removed (50.0%),
- Betweenness: 500 nodes removed (50.0%).

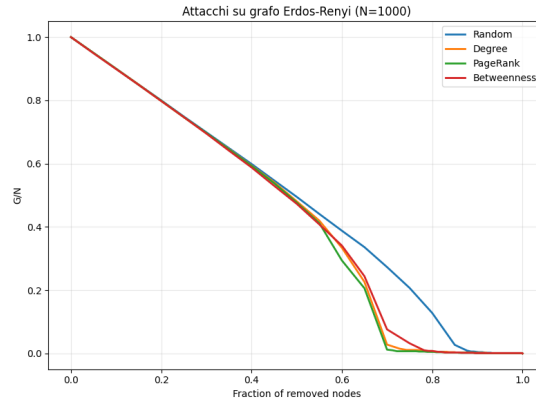


Figure 1: Attack results on the Erdős–Rényi graph. All curves collapse at 50%.

As shown in Figure 1, all attack strategies lead to collapse after removing exactly 50% of the nodes.

3.2 Barabási–Albert Network

The BA graph ($N = 1000$, $m = 5$) shows the typical behavior of scale-free networks. Random failures have a limited effect, while targeted attacks are more effective because they remove highly connected hubs.

Collapse points:

- Random: 500 nodes removed (50.0%),
- Degree: 375 nodes removed (37.5%),
- PageRank: 350 nodes removed (35.0%),
- Betweenness: 375 nodes removed (37.5%).

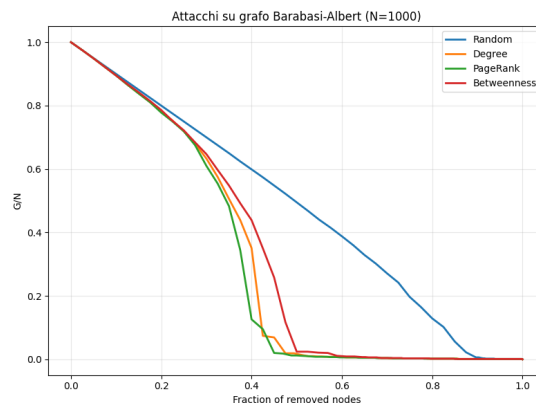


Figure 2: Attack results on the Barabási–Albert graph. Targeted attacks collapse the network earlier.

Figure 2 clearly shows that targeted attacks cause collapse significantly earlier than random removal.

4 Anime Similarity Network

The anime similarity network contains:

- 3579 nodes,
- 780393 edges,
- density = 0.1219,
- average degree ≈ 436.10 .

The network is dense and highly clustered, but not fully connected: there are 728 connected components. The giant component contains 2838 nodes (79.3% of the network).

Collapse points:

- Random: 1046 nodes removed (29.2%),
- Degree: 938 nodes removed (26.2%),
- PageRank: 1055 nodes removed (29.5%),
- Betweenness: 966 nodes removed (27.0%).

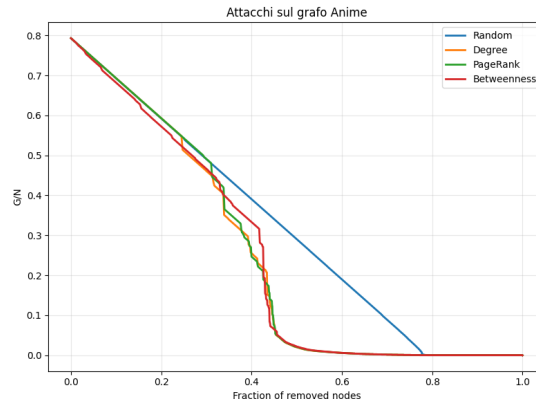


Figure 3: Attack results on the anime similarity network. Degree and betweenness attacks are the most damaging.

The collapse thresholds (fraction of removed nodes) are:

$$Random : 0.2923 \quad Degree : 0.2621 \quad PageRank : 0.2948 \quad Betweenness : 0.2699$$

PageRank does not outperform Degree or Betweenness because, in the anime similarity network, PageRank tends to select nodes that are highly connected within dense genre-based clusters. These nodes are not structurally critical for global connectivity. Degree and Betweenness attacks instead target hubs and bridging nodes, whose removal fragments the network more effectively.

5 Building Robustness

Three edge-addition strategies are applied:

1. **Ring Degree:** adds 600 new edges between neighbors of high-degree nodes.
2. **Ring Betweenness:** adds 1,487,668 new edges between neighbors of high-betweenness nodes.
3. **High Clustering:** adds 200 new edges between nodes with high clustering coefficient.

5.1 Critical Threshold

The theoretical critical threshold for random failures is:

$$f_c = 1 - \frac{1}{\frac{k_2}{k_1} - 1}$$

The results are:

Graph	N	L	k_1	k_2	f_c
Original	3579	780393	436.10	3.52×10^5	0.998758
Ring Degree (top-3)	3579	780993	436.43	3.52×10^5	0.998760
Ring Betweenness (top-3)	3579	2268061	1267.43	2.58×10^6	0.999508
High Clustering (200 edges)	3579	780593	436.21	3.52×10^5	0.998759

Table 1: Critical threshold comparison after robustness strategies.

Ring Degree and Ring Betweenness increase the theoretical threshold f_c by raising degree moments. High Clustering has negligible impact due to the small number of added edges.

6 Conclusion

The analysis shows that network robustness depends strongly on both structural properties and the attack strategy applied.

Random failures are generally less harmful because they do not specifically target structurally important nodes. This behavior is clearly visible in the Erdős–Rényi graph (Figure 1), where all attack strategies lead to collapse at the same point due to the absence of dominant hubs.

The Barabási–Albert graph (Figure 2) exhibits the expected scale-free behavior: it remains robust under random failures but becomes highly vulnerable when high-degree or high-centrality nodes are removed.

The anime similarity network (Figure 3) demonstrates high robustness thanks to its dense structure, large giant component, and strong redundancy. Targeted attacks are more effective than random removal, but the network maintains substantial resilience even when important nodes are removed.

The robustness-enhancing strategies modify the network in different ways. Ring Degree increases the connectivity of already central nodes, while Ring Betweenness provides a more balanced improvement by increasing redundancy around structurally important nodes. High Clustering has negligible impact because the number of added edges is extremely small compared to the original network size.

Overall, the results highlight how network topology shapes robustness and how targeted interventions can influence resilience in complex systems.